

CASE STUDY

The data existed. The problem was that no single layer could bring it together.

Regulated agricultural operations generate data at every stage: what was produced, what was packaged, what was sold, and at what rate. For one regional producer, that data existed across three separate systems, each accurate within its own domain. None of them connected to the others.

The question the operations team needed to answer, whether production was keeping pace with sales by product and strain, required pulling data from all three systems, reconciling naming differences by hand, and building a spreadsheet that was outdated before anyone read it. Armely replaced that process with a Microsoft Fabric Lakehouse that answers the question automatically.

About the client

A regional cultivation and processing company operating three licensed facilities with approximately \$42 million in annual revenue and 180 full-time employees. Operations are regulated at the state level with mandatory tracking through Metrc. Client details are generalized to protect confidentiality.

3 Source Systems Unified Google Sheets, Metrc, LeafLink	5 Power BI Reports Delivered Sales, production, inventory, customers, last order	0 Manual Exports Required After implementation
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The Problem

Three source systems held the data the operations team needed. Google Sheets tracked production volumes and rates. Metrc, the state-mandated tracking platform, held sales activity, facilities, packages, strains, and locations. LeafLink held order data, product categories, customer information, and rate-of-sales records.

Each system was maintained separately. None shared a common data model or an automated connection to the others. The practical consequence was a manual reporting process that required pulling data from each source, reconciling naming and categorization differences that varied by system, and assembling a consolidated view that was stale before it was finished.

The core operational question, whether a specific strain or SKU was being produced faster or slower than it was selling, could not be answered without a manual process that took time the team didn't have and produced a result that expired immediately.

The operations team had a red/yellow/green mental model for thinking about production-to-sales alignment. No automated equivalent existed. The threshold framework lived in someone's head, not in a system.

The data fragmentation problem across operations:

- Three systems with accurate data in each domain and no integration between them
- Manual reconciliation required across systems with different naming conventions and data structures
- Production-to-sales alignment invisible without a manual report rebuild each time
- Red/yellow/green thresholds existed conceptually but had no automated implementation
- No scalable path to add product categories or data sources without starting over

The Solution: Microsoft Fabric Lakehouse with Medallion Architecture

Armely assessed the environment and confirmed that none of the source systems needed to be replaced. Google Sheets, Metrc, and LeafLink each served their purpose. What was missing was the integration layer above them.

The solution was a Microsoft Fabric Lakehouse built on Medallion architecture: three structured layers that ingest, cleanse, and model data before delivering it to Power BI reports. Armely is an Authorized and Certified Metrc Integrator, which reduced the ramp time required to connect to and extract data from the state-mandated tracking system.

Bronze Layer: Data Ingestion

Fabric Data Factory pipelines were built to ingest data from all three sources into the Bronze layer. Google Sheets connected via a standard connector on a scheduled refresh. Metrc data was pulled via REST API, covering sales, facilities, deliveries, strains, packages, locations, and units of measure. LeafLink data was pulled via API, covering orders, products, product categories and subcategories, strains, product lines, facilities, and customers. Raw data landed in the Bronze layer unchanged.

Silver Layer: Cleansing and Standardization

The Silver layer resolved the naming and categorization differences that had made manual reconciliation necessary. Product names were standardized across systems. SKU alignment differences were resolved. The conceptual relationships between Google Sheets production data, Metrc inventory data, and LeafLink sales data were defined and encoded. The result was a consistent, structured dataset that the Gold layer could work with reliably.

Gold Layer: Data Modeling and Business Logic

The Gold layer built the logical relationships that supported slicing and filtering across customer, SKU, brand, category, strain, and date range. Business logic, aggregations, and calculations were applied at this layer, including the production-to-sales comparison calculations that power the red/yellow/green indicator system. Data was stored in Delta Tables for consistent query performance.

Five Power BI Reports

Five reports were built connected to the Gold layer, each covering a distinct operational domain.

- Sales Report and KPIs: current day, past day, past 7 days, month-to-date, past month, past year, and all-time views using LeafLink data.

- Production Orders: customer, SKU, brand, quantity, category, and discount views.
- Production Sold: profit/loss, units, and cases at the SKU and customer level.
- Customer Value: order counts and order totals by customer.
- Last Order: time since last order by customer, including license and total order history.

All reports include filtering, slicing, and sorting by date range, strain, and SKU.

Red/Yellow/Green Indicator System

Conditional formatting logic surfaces whether production is aligned, overproducing, or underproducing relative to sales, by product category, with drill-through to the strain and SKU level. Authorized users can adjust the threshold criteria directly, with all changes logged. A subsequent change order extended the system to apply thresholds beyond a one-week window using a multiplier on the par value logic.

Before	After
Production and sales in 3 systems with no integration	Unified Fabric Lakehouse with automated pipelines from all sources
Production-to-sales alignment required manual pull each time	Automated, available on demand with date-range filtering
SKU and naming differences required manual reconciliation	Standardized in Silver layer, consistent across all reports
Red/yellow/green thresholds informal, not automated	Configurable per category, logged, with drill-through to strain and SKU
Reports stale on arrival; manual rebuild required each time	Scheduled automated refresh across all 5 Power BI reports
No scalable path to add new sources or categories	Medallion architecture: new sources connect to existing Lakehouse

What Changed

The operations team answers the production-to-sales alignment question on demand, without a manual pull from three systems. The red/yellow/green dashboard surfaces the answer at a glance. Drilling into a specific strain or SKU takes one click.

The sales team sees customer order frequency, recency, and value without assembling a custom report. The Last Order report shows time since last order by customer with license and order history in one view.

The environment is documented and the client team was trained to administer it independently. When a new data source needs to connect or a new report needs to be built, the architecture is already in place to support it.

"The data fragmentation problem and the manual reporting burden it produces are present across agricultural production, food and beverage manufacturing, specialty chemicals, and

other regulated categories. The Microsoft Fabric platform and the Medallion architecture pattern can be applied to any organization with multiple operational data sources and a need for consolidated reporting."

— Armely, LLC

About Armely

Armely is a Dallas-based Microsoft solutions partner. Founded in 2017, we serve over 50 clients across energy, healthcare, government, and manufacturing. Our services cover Microsoft Dynamics 365, Microsoft Fabric, Power Platform, data analytics, and AI implementation. We work as an extension of your team, from initial strategy through deployment and ongoing support.

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